higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 A
1.1.2 B
1.1.3 B
1.1.4 A
1.1.5 D

(5 × 1) (5)

1.2
- Good strength
- Fatigue resistant
- Easy to fabricate
- High thermal conductivity
- Excellent ductility and toughness
- Corrosion resistant (Any 5 × 1) (5)

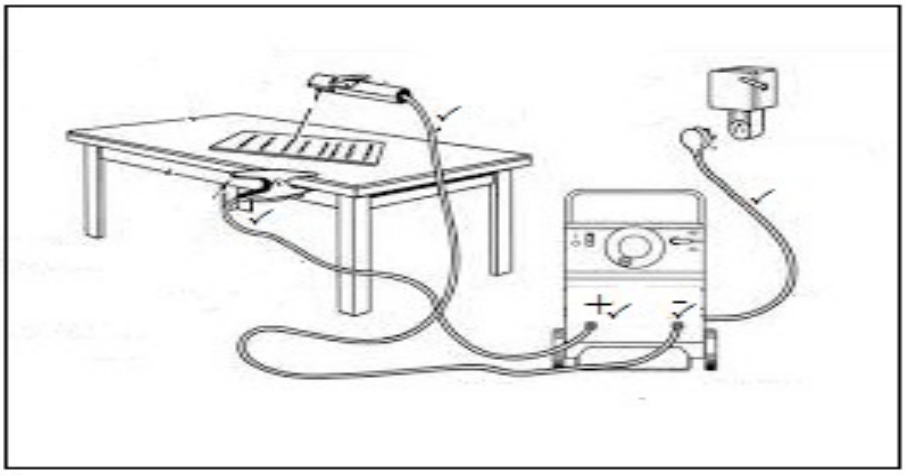
1.3 Connect the main power supply cable of the welding machine.✓ Connect the electrode holder cable to the positive fitting of the welding machine.✓Connect the earth clamp cable to the negative fitting of the welding machine.✓ Ensure that both cables have been secured. After completing the connecting process, switch the main power supply switch on and then the welding machine switch.✓ Set the amps according to the thickness of the electrode that is used for welding.✓ (5)

1.4
- Thickness of workpiece/plate
- Weld preparation
- Joint design
- Welding position
- Input heat/current (5)

[20]

QUESTION 2

2.1 Direct Current EARTH NEGATIVE



NOTE: + on work piece lead & - on electrode lead (5)

2.2 The reaction that takes place at the gas/liquid✓ metal interface, that is the splitting of oxygen,✓ nitrogen and hydrogen✓ into liquid metal under high temperatures✓ and the evolution of carbon dioxide.✓

OR

There is an arc✓ between the electrode tip and the workpiece. The arc generate heat✓ and melts✓ the base metal. The flux✓ coating on the electrode release a shielding✓ gas to protect the weld pool against atmospheric gasses. The flux also creates a cleaning✓ agent and removes all impurities from the molten metal forming a slag✓ on-top. Metal transfer happens inside the arc. ✓ (Any relevant 5 × 1) (5)

2.3
- Check the bevel angle to ensure good root penetration and fusion.
- Check the thickness of the pipe to determine the size of electrode.
- Check the welding position in which the pipe will be welded (horizontal, vertical or flat).
- Check and ensure that the pipe is clean and free from rust, paint and grease.
- Check the pipe is clamped correctly to the angle that it is to be welded. (5)

2.4
- To remove surface contamination
- To prevent unwanted chemical reaction
- To remove embedded iron that may cause stainless steel to rust
- To remove or repair mechanical damage or welding-related defects
- To assist in an inspection of the weld (5)

[20]

QUESTION 3

3.1 The welding process uses a non-consumable tungsten electrode to generate the weld.✓It is an arc welding process as it produces an electric arc between the tungsten electrode and the metal that is welded.✓ The inert gas, also called shielding gas, is usually argon or helium or a mixture of the two. The shielding gas protects the weld from atmospheric contamination.✓ The gas is fed through the welding torch. Welding begins when the tungsten electrode touches the metal. No slag is produced.✓ (4)

3.2 A – Shielding gas/Gas cylinder
B – Power source
C – Remote control/ foot pedal/amp setting
D – Filler Rod
E – Torch
F – Regulator/flow meter (6)

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3.3	3.3.1	D
	3.3.2	A
	3.3.3	J
	3.3.4	E
	3.3.5	K
	3.3.6	I
	3.3.7	B
	3.3.8	C
	3.3.9	H
	3.3.10	F

(10 × 1) (10)
[20]

QUESTION 4

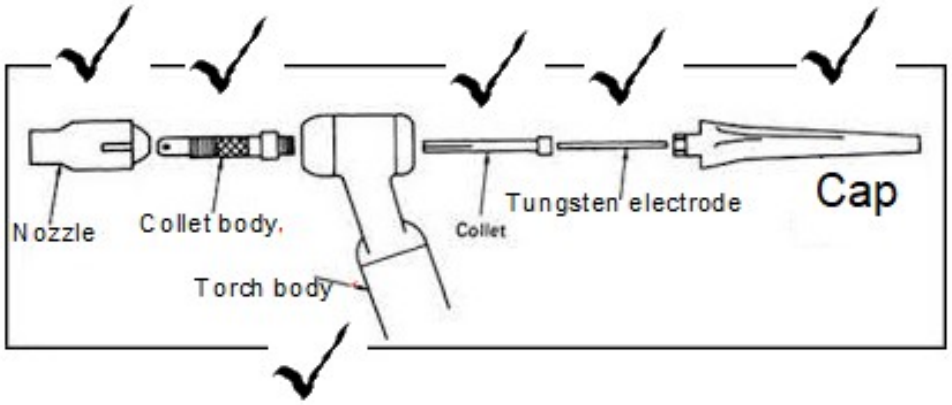
4.1
- SMAW electrodes are covered with a coating known as flux whereas tungsten electrodes do not have any flux coating.
- SMAW electrodes melt away during the welding process whereas tungsten electrodes do not melt that easily during welding.
- SMAW electrodes are cheaper than tungsten electrodes.
- SMAW electrodes have unique codes to indicate the type of electrode whereas tungsten electrodes are colour-coded.
- A very smooth weld is achieved when using tungsten electrodes whereas SMAW electrodes cause a lot of spatter during welding.

(Any relevant 2 × 2) (4)

4.2
- Incorrect temperature of the weld area
- Welding current too low
- Incorrect joint design
- Contamination
- Incorrect size of electrode (Any relevant 5 × 1) (5)

4.3 Welding is performed from the bottom of the joint.✓ The welder has to manipulate the weld pool quickly before it can fall into the torch nozzle.✓ The welder's wrist must move in a 'zigzag' motion.✓ The gas nozzle can rest on top of the joint and the arc is started in the centre of the joint.✓Pressure is applied to the nozzle once a pool is created. Progress the electrode and pause for a moment. Slide the right side of the nozzle forward so that it equals the left side.✓ (Any relevant 5 × 1) (5)

4.4



(6)
[20]

QUESTION 5

5.1
- Select the correct wire size.
- Set the correct gas flow on the regulator.
- Set the correct speed on the wire feeder machine.
- Set the correct current.
- Ensure that the wire matches the type of material that is to be welded.
- Clean the nozzle.
- Ensure that all connections are tight.
- Ensure that there is sufficient shielding gas.
- Check for gas leaks. (Any 5 × 1) (5)

5.2 The aluminium oxide coating✓ must be removed before welding can take place to prevent small traces of oxide from being trapped✓ in the weld.
The coating can cause a lack of fusion✓ resulting in cracking✓ of the weld.
Aluminium oxide absorbs moisture from the air and can cause porosity in the weld.✓ (5)

5.3
- Use a reamer
- Use a round file (2)

5.4
- Electrical power is delivered to the tip.
- The gun acts as a guide for the electrode wire.
- The gun delivers shielding gas to the weld area. (3)

5.5
- Gap or root opening
- Included angles
- Size of weld
- Weld pitch
- The parts of the weld that must be welded
- Length of weld

OR

- Weld all round
- Do not weld on site
- Fillet weld on arrow side
- Leg length of 6 mm
- Single bevel on the other side
- Single bevel with 45 degree (Any 5 × 1) (5)

[20]

TOTAL: 100

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